Stark Shifts in Simulated Vibrational Spectra

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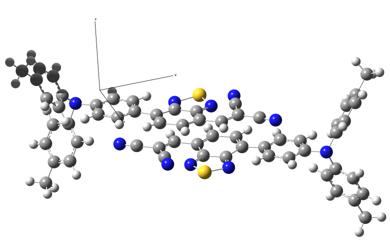
Abstract

We have modeled the effect of external electric fields on nitrile vibrational signatures of *2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3- benzothiadiazol-4-yl)methylene] propane-dinitrile* (DTDCPB). DTDCPB is a promising push-pull chromophore prevalent in organic photovoltaic research. By applying static electric fields *in silico* and simulating ground state IR spectra, we can calibrate the shifts with local electric fields. Born-Oppenheimer molecular dynamics was used to derive the model IR spectra to correlate to experimental data. In hopes of modeling pump-probe transient IR studies of DTDCPB, computational studies of excited-state dynamics are underway. These studies will enable *in operando* monitoring of electric fields of organic photovoltaic devices.

Graphical TOC Entry



The structure of the DTDCPB chromophore dimer under investigation in this work.[1]

Introduction

As research in solar energy becomes increasingly important, several new strategies are being investigated to increase the efficiency of solar devices. In organic photovoltaics (OPVs), one new approach uses small molecules with large ground-state dipole moments as electron donors. It is believed the inherent dipole allows for enhanced hole hopping and can minimize charge recombination.[2] One such molecule is 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene] propane-dinitrile (DTDCPB) and has resulted in OPVs with power conversion effiencies of up to 9.6%. The nitrile groups present in the molecule can act as reporter moieties for vibrational spectroscopy.

By modeling the effect of an applied electric field on absorption spectra (Stark Effect), it is possible to probe the local electric field during charge transfer events in a photovoltaic device via excited state IR spectroscopy. Understanding the mechanisms of charge transfer events enables rational design and enhancement of current materials for novel, high-efficiency semiconductor and photovoltaic devices. To perform these studies, we must first determine the electric field dependence of the ground and excited state IR spectra. The Stark effect can be modeled by frequency calculations in external static electric fields of various magnitudes in *Gaussian 09*. Spectral shifts obtained from these frequency calculations can be correlated to the presence of an electric field. Subsequently, Born-Oppenheimer Molecular Dynamics can be used to obtain model spectra using a different set of methods from the standard frequency calculations in *Gaussian 09* in order to compare to experimental data.[1]

Theory

Standard Frequency Calculations

The *normal modes of vibration* are calculated by diagonalizing a matrix of the second spatial derivatives of the electronic potential energy surface (PES) for the system of N atoms to yield the orthogonal set of N frequency normal modes $\{\omega_k\}$. The eigenvalues correspond to the normal modes of vibration for the system. The dipole moment μ is obtained from the instantaneous electron density around the nuclei. The intensities are derived from the spatial derivative of the dipole moment (Equation 1).[1]

$$\mathcal{A}_{\tilde{\nu}}(q_k) \propto \left(\frac{\partial \mu}{\partial q_k^2}\right) \tag{1}$$

The Stark Effect

The Stark effect is a phenomenon in which the energy level splittings are altered by the presence of an applied electric field. The effect is measured via changes to vibrational and optical spectra. The key parameters are the change in dipole moment ($\Delta\mu$) dependent on the degree of charge separation for a given transition, and a change in polarizability ($\Delta\alpha$) corresponding to the change in sensitivity of the vibrational transition to the applied field. The interaction of $\Delta\mu$ with an electric field is shown in Figure 1a.[3] The effect on an isotropic sample produces a broadened spectra (1b) and yields a differential absorbance spectra (1c) which follows the second derivative of the ground state absorption spectrum. Similarly, the effect of $\Delta\alpha$ is shown by Figure 1d-f. We can also model it computationally by performing calculations accounting for the presence of an external field. The change in frequencies is modeled by Equation 2 where $V(r_k)$ is the potential with respect to position r_k of electron k.[3]

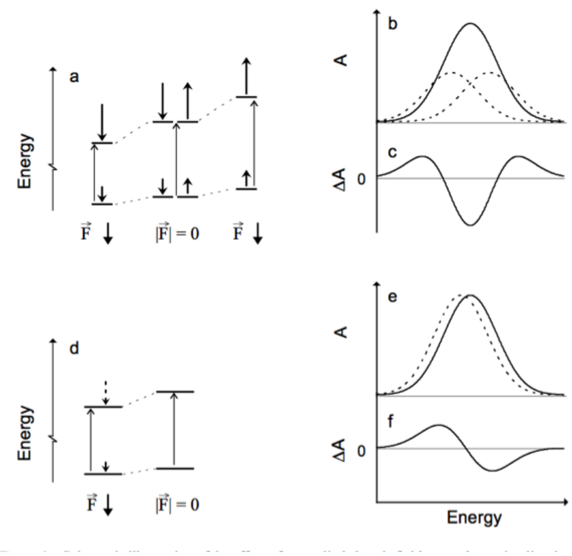


Figure 1: Illustration of the mechanism of the Stark Effect and the corresponding change in observed spectra.[3]

$$\Delta\omega_k = -\Delta\vec{\mu} \cdot \left(\frac{\partial V(\vec{r})}{\partial \vec{r}_k}\right) - \frac{1}{2}\Delta\alpha \cdot \left|\frac{\partial V(\vec{r})}{\partial \vec{r}_k}\right|^2 \tag{2}$$

Born-Oppenheimer MD (BOMD)

We apply Born-Oppenheimer Molecular Dynamics (BOMD) to derive the model spectra for DTDCPB with and without the static electric field present. BOMD utilizes Born-Oppenheimer approximate time-*independent ab initio* calculations to compute the electronic potential at each nuclear evolution time step. Integrating the classical equations of motion (EOMs) with the Velocity Verlet integration method (Equations 3) approximates nuclear dynamics.[1]

$$\begin{aligned}\vec{x}(t+\Delta t) &= \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2 \\ \vec{v}(t+\Delta t) &= \vec{v}(t) + \frac{\vec{a}(t) + \vec{a}(t+\Delta t)}{2}\Delta t.\end{aligned} \tag{3}$$

In this context, BOMD has advantages over the *ab initio* frequency calculation.[1,4] The Stark effect is most readily observed in vibrational spectra, so using purely *ab initio* techniques requires the second or third derivative of the electronic PES, which can be costly for larger systems. In contrast, BOMD only requires the gradient of the electronic PES to propagate nuclear position (Equations 3) and only the electron density to compute the time-dependent dipole moment (Equation 4). Thus, BOMD manages to capture some anharmonicity and higher order terms of the potential $V(\vec{r}_k)$ using only the gradient for each time step.[1]

$$\vec{\mu}(t) = -\sum_{mu}\sum_{nu} P_{\mu\nu}\big(\nu|\mathbf{r}(t)|\mu\big) + \sum_{A} Z_A \mathbf{R}_A(t) \tag{4}$$

Obtaining Spectra from Dipole Orientation Time Series

Since BOMD only calculates the first derivative of the PES, another method is needed to obtain spectra. The dipole vector $\vec{\mu}(t)$ is output from BOMD as a time series, so taking the first time derivative $\dot{\vec{\mu}}(t)$ and using the *dipole autocorrelation function* $\langle \dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t+\tau)\rangle_\tau$ as the kernel of an integral transform (Equation 5) we obtain the normal mode frequencies and intensities.[4]

$$\mathcal{A}(\omega) \propto \int \langle \dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t+\tau)\rangle_\tau e^{-i\omega t} dt. \tag{5}$$

Methodology

All *ab initio* calculations performed in this work used the B3LYP/6-311G(d,p) level of theory. The static field was chosen rather than an oscillating field to simulate the local environment around each DTDCPB dimer immediately after another nearby dimer was stimulated into a charge transfer state. The standard frequency calculations for the ground state of DTDCPB both with and without an applied static field were performed to determine the equilibrium geometries in the various environments and generate IR spectra. The field was generated from a static dipole oriented along the z-axis normal to the plane in which the majority of the DTDCPB molecule was embedded.

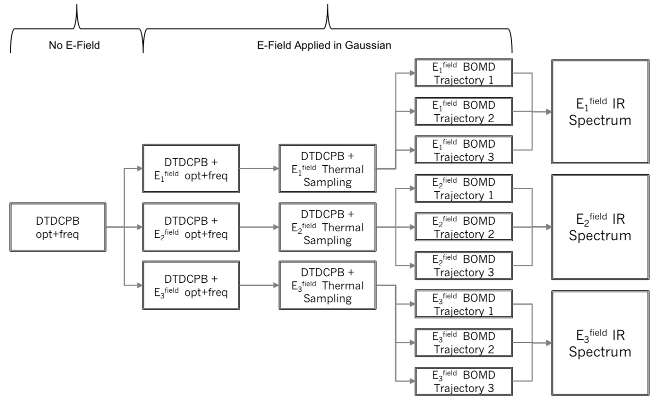


Figure 2: Diagram of the sequence of calculations for obtaining simulated spectra with BOMD. The process is identical for both the cyanide test cases and the DTDCPB monomer.

BOMD was applied to a cyanide ion as a test case to model the nitriles on DTDCPB. Subsequently, it was applied to the monomer in zero field and applied fields. The systems were brought to ideal initial conditions "Production MD" steps by performing thermal Monte Carlo sampling in the electric field to generate multiple nuclear trajectories with thermally populated vibrational normal modes. The production MD runs yielded the dipole time series for calculating the spectra. The individual spectra from the multiple production MD runs can be averaged in the frequency domain to obtain a single spectrum from multiple BOMD runs. Our BOMD application scheme follows that outlined in Figure 2.

Computational Results

Two nitrile stretching frequencies were detected through the calculation, which will be referred to as the "symmetric" and "asymmetric" stretches. Their frequencies were 2356.48 and 2334.28 cm^{-1}, respectively. Figure 3 shows the displacement vectors of both stretching frequencies, where in the symmetric stretch both nitrogens stretch in unison and in the asymmetric stretch the nitrogens move in opposition to one another.

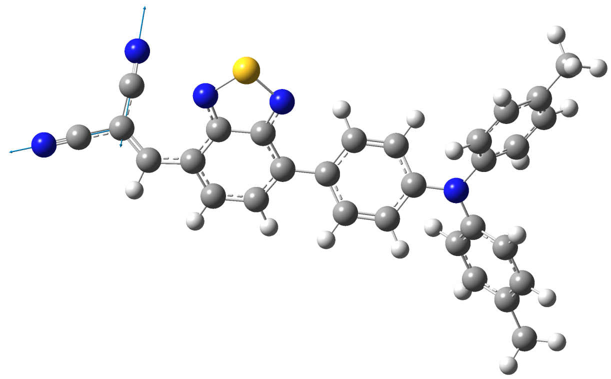


Figure 3: Structure of ground state DTDCPB, indicating displacement vectors of nitrile stretching frequency.

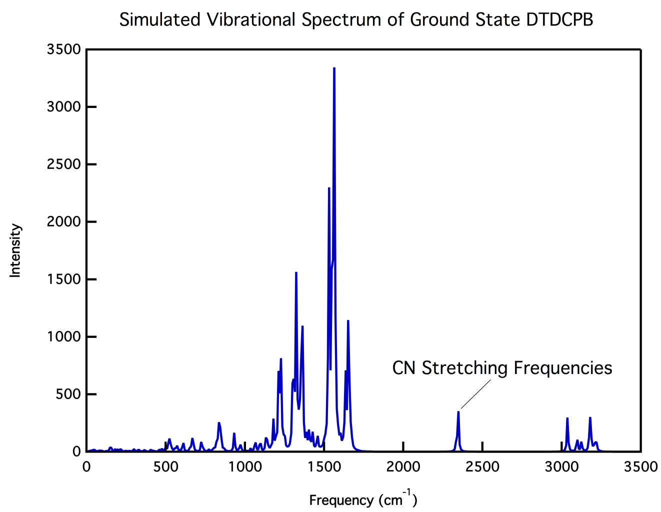


Figure 4: Ground state IR spectrum of DTDCPB.

The simulated vibrational spectrum is shown in Figure 4. Although both CN stretching frequencies were calculated, the intensity of the asymmetric stretch was very low, so more focus was placed on the symmetric stretch which had the higher intensity which can be seen in the figure. Experimental IR data would be needed to conclude that the symmetric stretch is of more relevant focus.

The frequencies of both stretches in the presence of a static field are presented below. It is important to note that an applied field of 50 MV/cm is approximately the field strength generated in hydrogen bonding and that of 25 MV/cm is representative of dipole-dipole interactions.

Applied Field (MV/cm)	**Symmetric Stretch (cm^{-1})**	**Asymmetric Stretch (cm^{-1})**
50	2331.56	2316.55
25	2338.33	2326.76
0	2346.48	2334.28
-25	2338.97	2328.01
-50	2331.21	2318.22

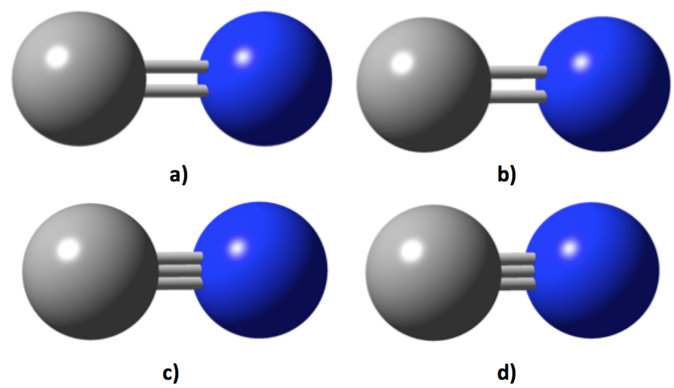


Figure 5: Four orientations of the CN group along the BOMD trajectory. The bond lengths at each point are a) 1.277 Å, b) 1.234 Å, c) 1.156 Å, and d) 1.094 Å.

BOMD for a cyanide group was executed as per the Methodology with success. Figure 5 shows the cyanide group at various bond lengths along its trajectory of vibration. The average bond length along the trajectory was determined to be 1.189 Å, which is in fair agreement to experimental data for hydrogen cyanide that has bond length of 1.156 Å. To determine if the BOMD process was executed correctly, the energy of the system was monitored as a function of time, as shown in Figure 6. Total energy is conserved and kinetic and potential energy oscillate as one would expect for this two particle system.

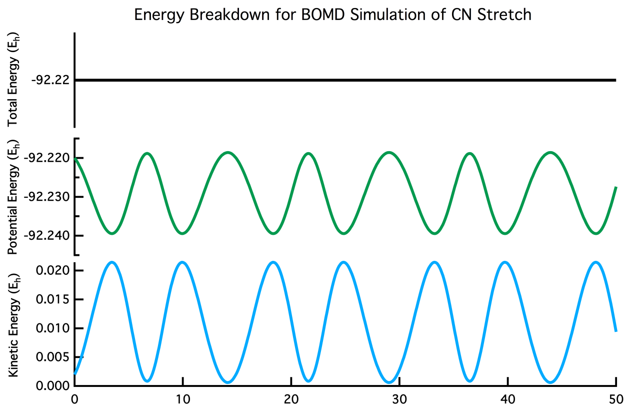


Figure 6: Plots of the kinetic and potential energy along the BOMD timesteps of cyanide are shown in green and blue. Total energy, shown in black, is conserved over the course of the trajectory.

Conclusion

From the computational results, there appears to be a Stark shift occurring in the ground state spectra which is nearly symmetric with respect to inversion of applied field orientation in the case of the symmetric stretch. This would indicate that polarizability is the dominant factor behind this Stark shift, as only the magnitude of the field influences the change in frequency. Although the asymmetric stretch shows a Stark shift, its low intensity leads us to believe it would not affect the Stark spectrum of DTDCPB as a whole. Furthermore, we found that BOMD could be applied to cyanide and maintain proper energy conservation in the zero field case. The average bond length determined for this test case was also in agreement with experimental data. After performing BOMD on cyanide in an applied field, we can extend this work to study the entire DTDCPB dimer system with and without applied fields in the ground state, which forms the basis for our continuing work. Being able to computationally model ground and excited state dynamics would allow us to fully simulate the charge transfer events present in this important organic photovoltaic.

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